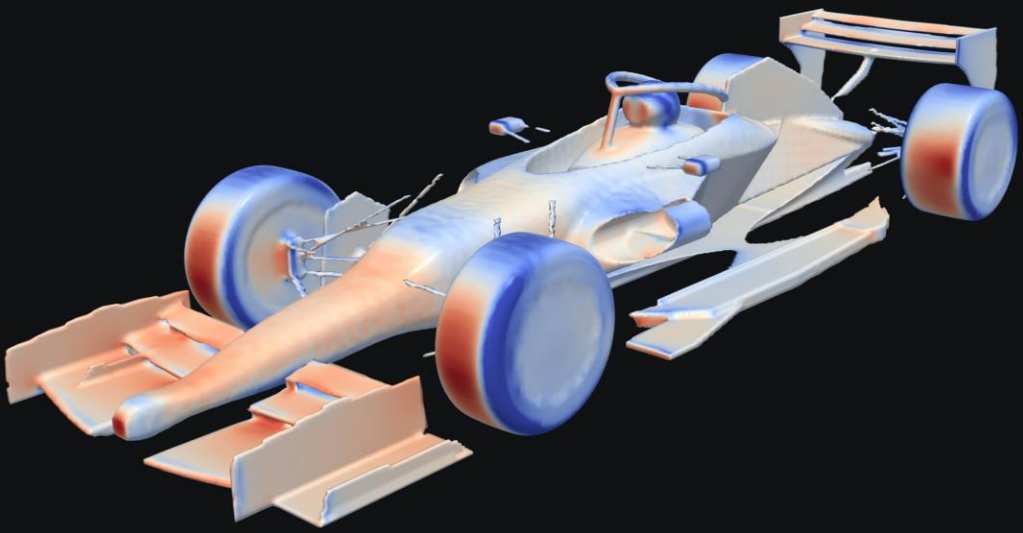
Where does a 41% downforce difference come from?

Near-wall mesh sensitivity study on a generic 2026 F1 model



Surface pressure coefficient Cp (layered mesh)

−1.0 −0.5 0.0 0.5 1.0

Same case, two meshes. Drag moved 0.6%, downforce 41%.

95% of that difference sits on the rear tyres.

Setup

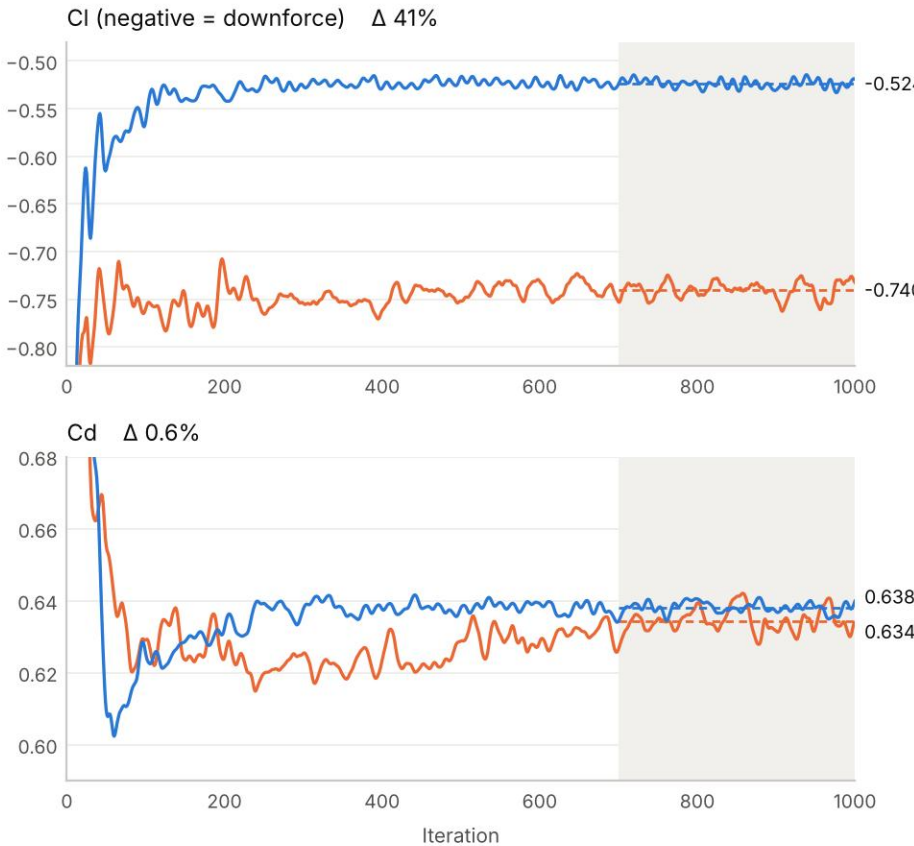
Everything identical between the two cases except the near-wall mesh.

Geometry	Generic 2026 F1 STL, repaired (424 self-intersections)
Solver	OpenFOAM v2512, simpleFoam (steady RANS)
Turbulence	k-ω SST with wall functions
Inlet turbulence	k 0.24 m^2/s^2, ω 1.78 1/s (tutorial values)
Flow	20 m/s inlet, moving ground at 20 m/s
Wheels	Static (no-slip), 20 mm gap to the road
Domain	20 × 8 × 6 m, symmetry top/sides, 2.8% blockage
Mesh A	snappyHexMesh, no prism layers
Mesh B	Same base mesh + 1 prism layer, 88% coverage
Reference area	1.327 m^2, frontal area from the STL
Statistics	Mean of iterations 700–1000 (of 1000)

Drag barely moves. Downforce does.

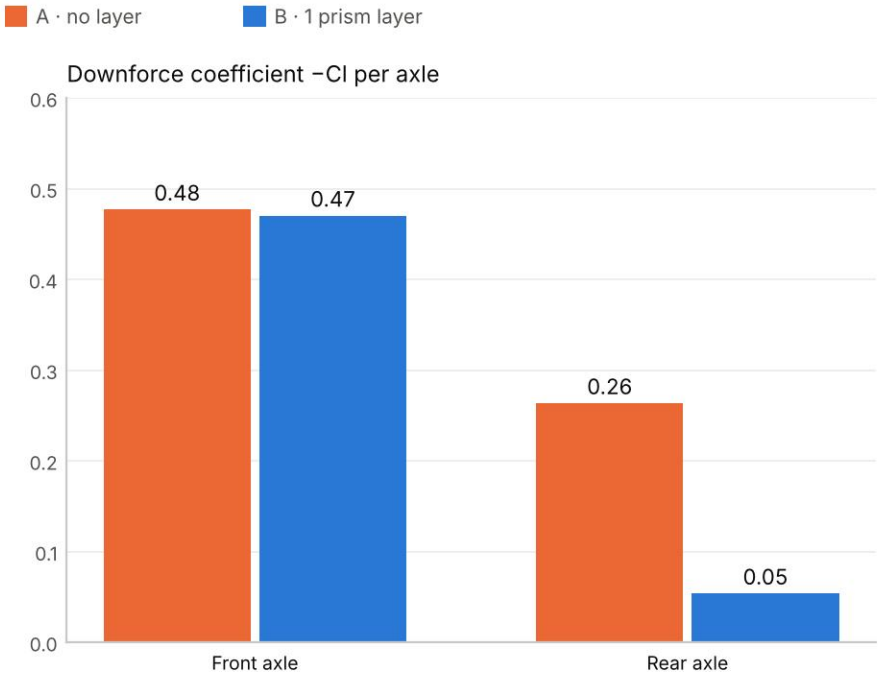
Force coefficients vs iteration. Shaded: averaging window.

A · no layer B · 1 prism layer



The front axle doesn't change

Downforce per axle from the pitching moment, taken about the tyre contact centres (x = 0 and 3.40 m).

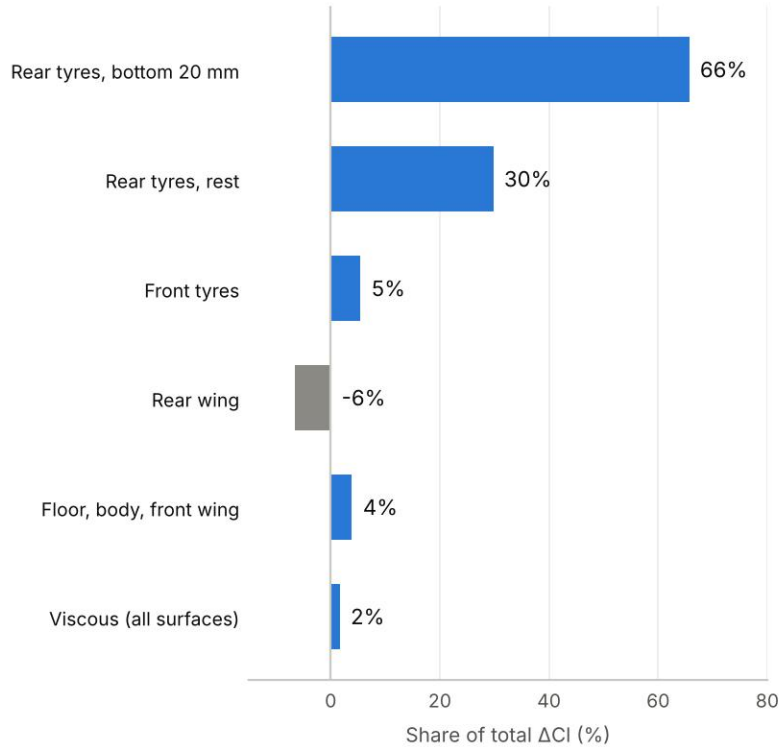


Front share of downforce: 64% (A) vs 90% (B).

Compare the meshes, not the absolute balance: static wheels and a generic geometry put this far from a real car (L/D ≈ 0.8 on mesh B).

Where the difference comes from

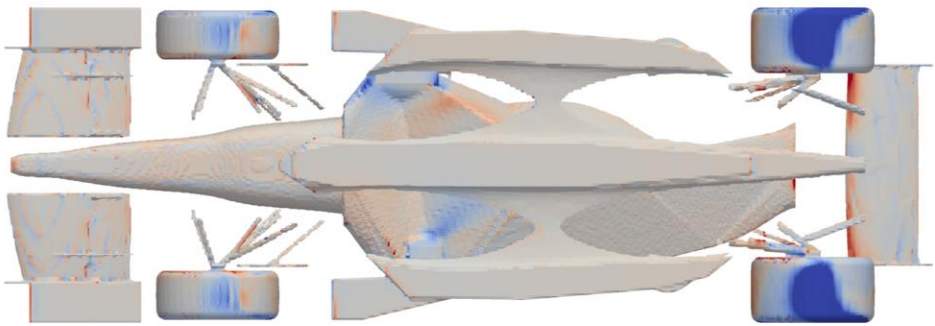
Pressure difference (A − B) integrated over each region, as a share of the total change in Cl. Viscous part from the force totals.



The rear tyres carry 95% of it, and the bottom 20 mm of the rear tyres alone 66%. Wings, floor and body barely move.

Pressure difference, underside

ΔCp = Cp(A, no layer) − Cp(B, layered). Blue: lower pressure (more suction) on the no-layer mesh.



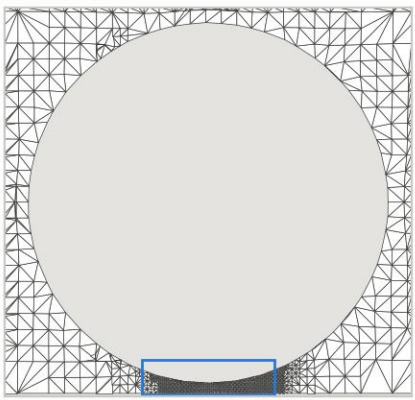


View from below, nose on the left. The signal is concentrated where the rear tyres sit above the road.

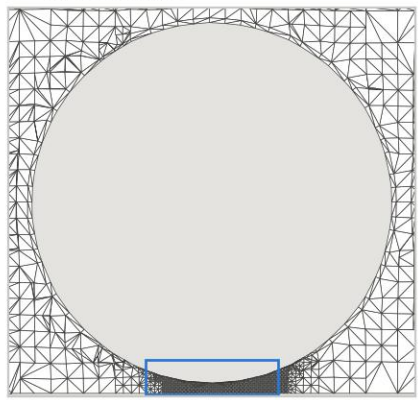
The 20 mm tyre–road gap

Cut through the right rear tyre (y = 0.75 m). The tyre sits 20 mm above the moving road so the mesh stays valid.

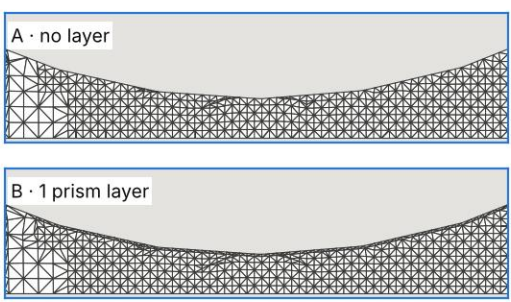
A · no layer



B · 1 prism layer



Detail: the gap under the tyre (blue box), 26 cm wide

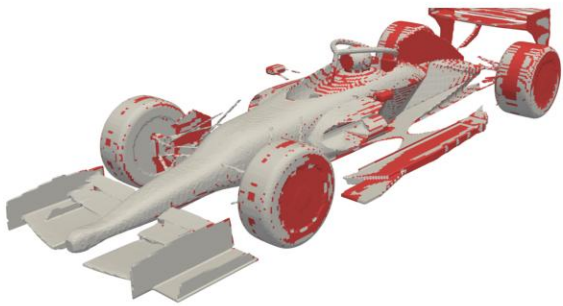


y+: what the layer does fix

Red: y+ above 300, outside the log-law range wall functions assume. The layer fixes this almost everywhere, yet the wings and floor forces barely change (slide 5).

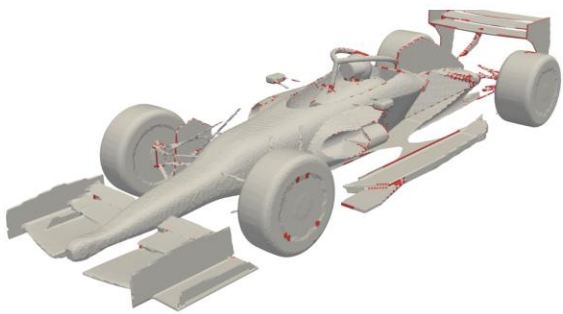
A · no layer

30.0% of surface faces above y+ 300 · median y+ 231



B · 1 prism layer

4.0% of faces above y+ 300 · 1.1% below 11 · median y+ 98

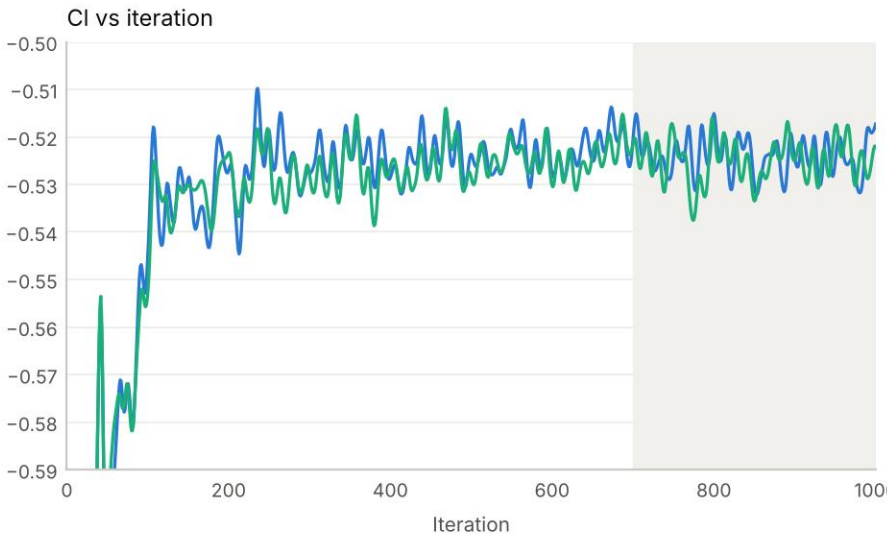


y+ > 300 11–300 y+ < 11

1 core vs 8 cores

Mesh B run twice from the same initial conditions: serial and 8-way scotch decomposition.

Serial 8-way parallel



	Serial	8-way parallel	Δ / σ
Cd	0.6380 ± 0.0018	0.6387 ± 0.0016	0.27
Cl	-0.5239 ± 0.0039	-0.5250 ± 0.0043	0.18

Both differences are well under one standard deviation of the run's own oscillation. The decomposition doesn't change the answer.

Before trusting absolute numbers

What I'd change next, in order of expected impact.

1 **Model the contact patch**

Replace the 20 mm gap with a plinth / flattened contact patch.

2 **Rotate the wheels**

Split tyres and rims into own patches; rotating-wall BC on tyres.

3 **Resolve the near-wall region properly**

3–5 prism layers, target y+ 30–100 on the wings and floor.

4 **Mesh independence**

Three refinement levels and a grid-convergence estimate.

5 **Run at 50 m/s**

The FIA wind-tunnel speed cap; also raises Reynolds number.

6 **Then: wake survey**

Split drag into induced and profile parts from wake planes.